1 Experimental Code

1.1 Experiment 1: Lexical Pitch Accent Classification

```
1
2  // This is run with jsPsych 6.3. These external JS/CSS files were
       loaded in:
3  // jspsych-html-keyboard-response.js
4  // jspsych-instructions.js
5  // jspsych-preload.js
6  // jspsych-audio-tokens-v7.js
7  // jspsych-audio-tokens.js
8  // jspsych-html-button-response.js
9  // d3.js
10 // audio-tokens.js
11
12
13 var timeline = [];
14
15 var preload = {
16     type: 'preload',
17     auto_preload: true
18 }
19 timeline.push(preload);
20
21 var Welcome = {
22    type: 'html-button-response',
23    stimulus: '<p>Welcome to the demo of the experiment!</p>',
24    choices: ['Begin']
25 }
26
27 timeline.push(Welcome);
28
29 /////////////// Experiment ///////////
30
31 // Add all trials to a timeline array
32
33 var stimuli2 = [
34     'FKavan_LR.mp3',
35     'MKavan_SR.mp3',
36     'FKavan_LF.mp3',
37     'MKavan_SF.mp3',
38     'FRejav_SR.mp3',
39     'MRejav_LF.mp3',
40     'FRejav_SF.mp3',
41     'MRejav_LR.mp3'
42 ]
```

```
43
44  var instructions1 = {
45      type: 'instructions',
46      pages: [
47          `<p>You will see a big hollow circle and eight smaller circles
48          around it. Hover over each circle to listen to it.</p>`,
49      ],
50      show_clickable_nav: true,
51      response_ends_trial: true
52  }
53
54  var instructions2 = {
55      type: 'instructions',
56      pages: [
57          `<p>Listen and group the items based on their similarity.
58          Simply drag the circles inside. The more similar, the closer
59          they should be.</p>`,
60      ],
61      show_clickable_nav: true,
62      response_ends_trial: true
63  }
64
65  var similarity_testing1 = {
66      type: 'audio-tokens',
67      ratingtype: 'similarity',
68      stimuli: stimuli2,
69      force_listen: true,
70      loop: false,
71      draw_edges: true
72  }
73
74  // Add all trials to a timeline array
75  var good2 = {
76      timeline: [
77          instructions1,
78          instructions2,
79          similarity_testing1
80      ]
81  };
82
83  timeline.push(good2);
84
85  var ending = {
86      type: 'html-button-response',
87      stimulus: `Thank you, the demo has been completed.<br><br>`,
88      choices: ['Press here to finish'],
89      on_finish: function() {
90          jsPsych.endExperiment(`You may close this page. Thank you!`);
91      },
92  };
93
94  timeline.push(ending);
95
96  jsPsych.init({
```

```
    timeline: timeline,
    show_preload_progress_bar: true,
    message_progress_bar: `My progress`,
    show_progress_bar: true,
    use_webaudio: false,
    case_sensitive_responses: true,
})
```

1.2 Experiment 2: Vowel Free-Classification Task

```
// This is one of the pre-test tasks. The exact combination of voices used as the stimuli depended on which participant was doing the task,
// so that all voices were counterbalanced.

// This is run with jsPsych 6.3. These external JS/CSS files were loaded in:
// jspsych-html-keyboard-response.js
// jspsych-instructions.js
// jspsych-preload.js
// jspsych-audio-tokens-v7.js
// jspsych-audio-tokens.js
// jspsych-html-button-response.js
// d3.js
// audio-tokens.js

var timeline = [];

var preload = {
    type: 'preload',
    auto_preload: true
}
timeline.push(preload);

////////////// PARTICIPANT INSTRUCTIONS //////////////

var Instructions = {
    type: 'html-button-response',
    stimulus:
        '<p>You will see a circle with many bubbles around it. ' +
        'When you mouse over a bubble, an Anatolian word will play. ' +
        'You can drag and drop the bubbles into the circle. ' +
        'Your task is to organize the words according to their vowels. </p>' +
        '<p>Please ensure that your volume is turned up.</p>' +
        '<p>Press `begin` to begin.</p>',
    choices: ['Begin']
}

timeline.push(Instructions);

////////////// CRITICAL TASK //////////////

// Circle 1

var stimuli1 = [
    'voice2_fyZ_61.wav',
    'voice2_fiZ_61.wav',
    'voice2_fuZ_61.wav',
    'voice2_fWZ_61.wav',
```

```
    'voice2_bys_61.wav',
    'voice2_bis_61.wav',
    'voice2_bus_61.wav',
    'voice2_bWs_61.wav',
    'voice2_dyf_61.wav',
    'voice2_dif_61.wav',
    'voice2_duf_61.wav',
    'voice2_dWf_61.wav',
    'voice2_vyS_61.wav',
    'voice2_viS_61.wav',
    'voice2_vuS_61.wav',
    'voice2_vWS_61.wav'
]

var similarity_testing1 = {
    type: 'audio-tokens',
    ratingtype: 'similarity',
    stimuli: stimuli1,
    force_listen: false,
    loop: false,
    draw_edges: true
}

var good1 = {
    timeline: [
        similarity_testing1
    ]
};

timeline.push(good1);

// Circle 2

var stimuli2 = [
    'voice2_syf_61.wav',
    'voice2_sif_61.wav',
    'voice2_suf_61.wav',
    'voice2_sWf_61.wav',
    'voice2_zyg_61.wav',
    'voice2_zig_61.wav',
    'voice2_zug_61.wav',
    'voice2_zWg_61.wav',
    'voice2_tSyv_61.wav',
    'voice2_tSiv_61.wav',
    'voice2_tSuv_61.wav',
    'voice2_tSWv_61.wav',
    'voice2_nym_61.wav',
    'voice2_nim_61.wav',
    'voice2_num_61.wav',
    'voice2_nWm_61.wav'
]

var similarity_testing2 = {
    type: 'audio-tokens',
```

```
    ratingtype: 'similarity',
    stimuli: stimuli2,
    force_listen: false,
    loop: false,
    draw_edges: true
}

var good2 = {
    timeline: [
        similarity_testing2
    ]
};

timeline.push(good2);

////////////// FILLER TASK ////////////

var Instructions2 = {
    type: 'html-button-response',
    stimulus:
        '<p>You will see a circle with many bubbles around it. ' +
        'When you mouse over a bubble, an English word will play. ' +
        'You can drag and drop the bubbles into the circle. ' +
        'Your task is to organize the words according to their
           meanings.</p>' +
        '<p>Press `begin` to begin.</p>',
    choices: ['Begin']
}

timeline.push(Instructions2);

// Circle 1

var stimuli3 = [
    'charys_tomato.wav',
    'charys_potato.wav',
    'charys_rain.wav',
    'charys_plate.wav',
    'charys_dolphin.wav',
    'charys_octopus.wav',
    'charys_penguin.wav',
    'charys_bicycle.wav',
    'charys_passport.wav',
    'charys_airplane.wav',
    'charys_forest.wav',
    'charys_bookstore.wav',
    'charys_library.wav',
    'charys_mirror.wav',
    'charys_calendar.wav',
    'charys_museum.wav'
]

var similarity_testing3 = {
    type: 'audio-tokens',
```

```
    ratingtype: 'similarity',
    stimuli: stimuli3,
    force_listen: false,
    loop: false,
    draw_edges: true
}

var good3 = {
    timeline: [
        similarity_testing3
    ]
};

timeline.push(good3);

// Circle 2

var stimuli4 = [
    'charys_photo.wav',
    'charys_castle.wav',
    'charys_lawyer.wav',
    'charys_paper.wav',
    'charys_desk.wav',
    'charys_coin.wav',
    'charys_oasis.wav',
    'charys_desert.wav',
    'charys_river.wav',
    'charys_hospital.wav',
    'charys_doctor.wav',
    'charys_microscope.wav',
    'charys_rocket.wav',
    'charys_planet.wav',
    'charys_fork.wav',
    'charys_bottle.wav'
]

var similarity_testing4 = {
    type: 'audio-tokens',
    ratingtype: 'similarity',
    stimuli: stimuli4,
    force_listen: false,
    loop: false,
    draw_edges: true
}

var good4 = {
    timeline: [
        similarity_testing4
    ]
};

timeline.push(good4);

////////////// END EXPERIMENT //////////////
```

```
209
210 var block = {
211     type: 'html-keyboard-response',
212     choices: ['a', 'A'],
213     stimulus: '<p>Press [a] to complete the experiment.</p>',
214     stimulus_width: 300,
215     data: {
216         correct_response: 'a',
217     },
218     on_finish: function(data) {
219         if (data.response.toLowerCase() == data.correct_response) {
220             jsPsych.endExperiment(
221                 'Thank you for participating! Please proceed to Part 2
                        ' +
222                 '<a href=`https://sarcastirony.ca/students/preAX/' +
223                 'pretest-subject33-speaker2.html`>here</a>.'
224             );
225         }
226     }
227 };
228
229 timeline.push(block);
230
231 jsPsych.init({
232     timeline: timeline,
233     show_preload_progress_bar: true,
234     message_progress_bar: `My progress`,
235     show_progress_bar: true,
236     use_webaudio: false,
237     case_sensitive_responses: true,
238 })
```